AREA CHARTS AND RELATED VISUALIZATIONS IN POWER BI

Data visualization is an essential component of business intelligence, allowing analysts to understand trends, compare measures, and uncover patterns that are not immediately visible in raw data. Power BI provides a wide variety of visualizations, each serving a specific analytical purpose. Among these, area charts, stacked area charts, ribbon charts, waterfall charts, funnel charts, and scatter charts are frequently used to analyze time-based patterns, categorical summaries, and relationships between variables. This document provides comprehensive notes on these visualizations and their formatting options based on step-by-step operations performed in Power BI.

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1. AREA CHART

1.1 Purpose of Area Chart

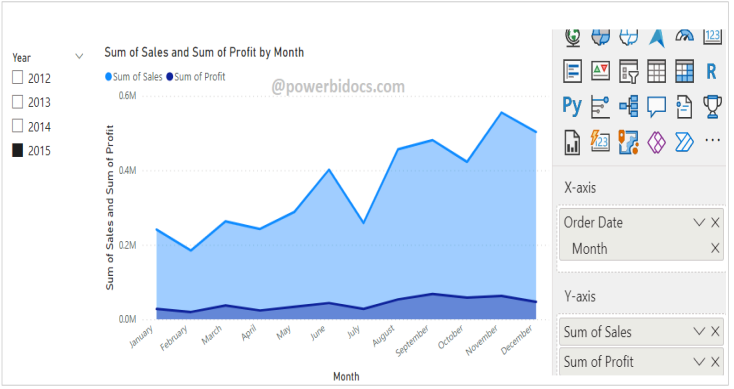
An area chart is an extension of a line chart where the area between the line and the axis is filled with a color. It is especially useful when:

- You want to analyze trends over time.
- You have one or more continuous numeric variables.
- You want to emphasize the magnitude of change.

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1.2 Creating an Area Chart

1. Select the **Area Chart** visual from the visualization pane in Power BI.
2. Increase the chart template size for better visibility.
3. Drag the **Date** field to the **X-axis**.
    - In Power BI, date fields are often auto-converted into hierarchy (Year → Quarter → Month → Day).



    - Switch from *Date Hierarchy* to *Date* (exact date) for a continuous timeline.

4. Drag continuous numerical fields (e.g., **Sales** and **Profit**) to the **Y-axis**.

1.3 Why Add Multiple Variables?

A single variable area chart looks similar to a line chart with shading.
Adding multiple continuous measures (e.g., Profit and Sales):

- Highlights differences in magnitude.
- Makes it easier to compare trends.
- Helps identify periods of positive and negative performance (e.g., negative profit).

1.4 Interpretation

- **Light Blue Area** → Sales
- **Purple / Violet Area** → Profit
- Regions where profit dips below zero show a negative performance.
- Alignment between rising/falling sales and profit indicates correlation.
- Misalignment (e.g., high sales but negative profit) signals inefficiencies or costly sales events.

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1.5 Using Zoom Slider

Power BI enables zooming on both X-axis (Date) and Y-axis:

- Helps analyze specific periods (e.g., February only).
- Allows analysts to compare peaks and troughs precisely.
- Useful for focusing on a particular time window without filtering the entire report.

1.6 Enhancing the Area Chart

You can format the chart using:

- **Titles** (Chart Title, X-axis Title, Y-axis Title)
- **Gridlines** (Horizontal + Vertical)

- **Transparency adjustments**
- **Line smoothness settings**
- **Data Labels** (size, color, boldness)
- **Series Labels** (to represent legend information directly on the chart)
- **Plot area background images**

Adding details in the “Details” well allows including extra numerical fields such as:

- **Quantity**
- **Customer age**
- **Satisfaction score**, etc.

This helps you hover over a data point and see a more enriched insight.

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1.7 Analytical Insights from Area Charts

Example interpretation:

- On Wednesday, Sales = 707 and Profit = 168.
- On Friday, Sales = 707 but Profit = -25 (negative).
  This difference prompts deeper investigation:
- Which products were sold on both days?
- Were discounts applied?
- Was Friday’s sales mix less profitable?

Thus, area charts support decision-making by identifying abnormal financial behavior.

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2. STACKED AREA CHART

2.1 Understanding Stacked Area Chart

A stacked area chart superimposes multiple area layers on top of one another.
Unlike a basic area chart:

- The values are shown cumulatively.
- The chart helps show how individual components contribute to the total.

2.2 Key Differences

Feature	Area Chart	Stacked Area Chart
Comparison	Absolute trend comparison	Contribution comparison
Values	Independent	Cumulative
Use-case	Track individual performance	Show total and component breakdown

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2.3 When to Use

- For total sales and total profit visualization.
- When analyzing different categories contributing to total over time.
- When overall magnitude matters.

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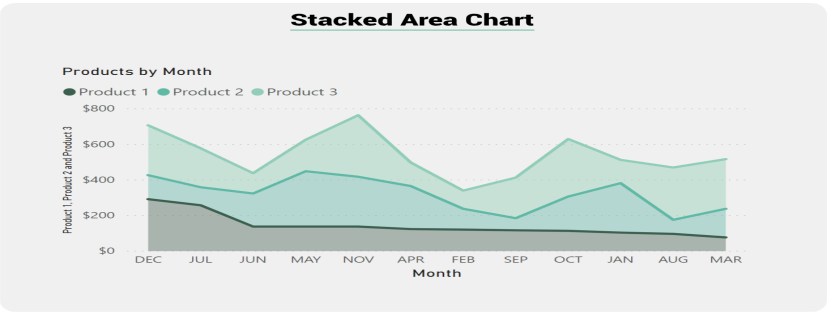
3. 100% STACKED AREA CHART

3.1 Purpose

A 100% Stacked Area Chart represents the proportion of each variable relative to the total, standardized to a 100% scale.

3.2 Use Cases

- When the intent is to compare relative percentages rather than



absolute values.

- To understand which variable dominates on which date.

3.3 Interpretation

- Dark areas may indicate negative values.
- Useful when number ranges differ significantly but comparative significance matters.

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4. COMBO CHARTS (LINE + COLUMN)

4.1 Line and Stacked Column Chart

This chart:

- Represents one measure as a line (e.g., Profit).
- Represents another measure as stacked columns (e.g., Sales).
- Enables comparison between two different types of trends.

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4.2 Line and Clustered Column Chart

Clustered columns display side-by-side bars for comparison of categories.
Combined with a line:

- Helps compare category groups against trends.

4.3 Features

- Zoom slider for readability.
- Color customization for both line and columns.
- Data labels available for both components.
- Adding Column Legend (if applicable).

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5. RIBBON CHART

5.1 Overview

A ribbon chart is used to display ranking changes over time:

- The width and position of ribbons indicate magnitude and movement.
- It effectively shows how categories rise or fall in importance across periods.

5.2 Creating Ribbon Chart

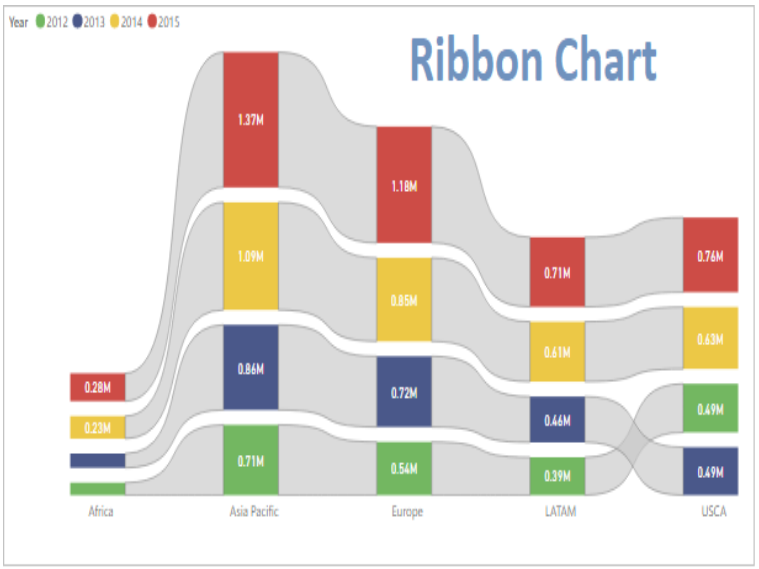
Use the same fields used in line + column chart:

- Date on X-axis
- Sales and Profit measures
- The chart visually transitions into layered ribbons when switched.

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5.3 Interpretation

- Light bands indicate Sales.
- Dark bands indicate Profit.
- Helps identify whether categories maintain consistent ranking.



5.4 Application Areas

- Sales performance ranking
- Market share trends
- Profit movement analysis
  Ribbon chart provides a visually appealing, trend-focused layout.

6. WATERFALL CHART

6.1 Characteristics

A waterfall chart:

- Shows how individual category values add up to a final total.
- Visually resembles cascading bars.

6.2 Use Cases

- To explain how total revenue is constructed from category-level contributions.
- Financial statements (revenue → expenses → net profit).
- Inventory changes.

6.3 Steps

1. Place a categorical hierarchy (e.g., Category → Subcategory) on the X-axis.
2. Use a numerical variable like **Quantity**.
3. Power BI displays incremental additions leading to a final total.

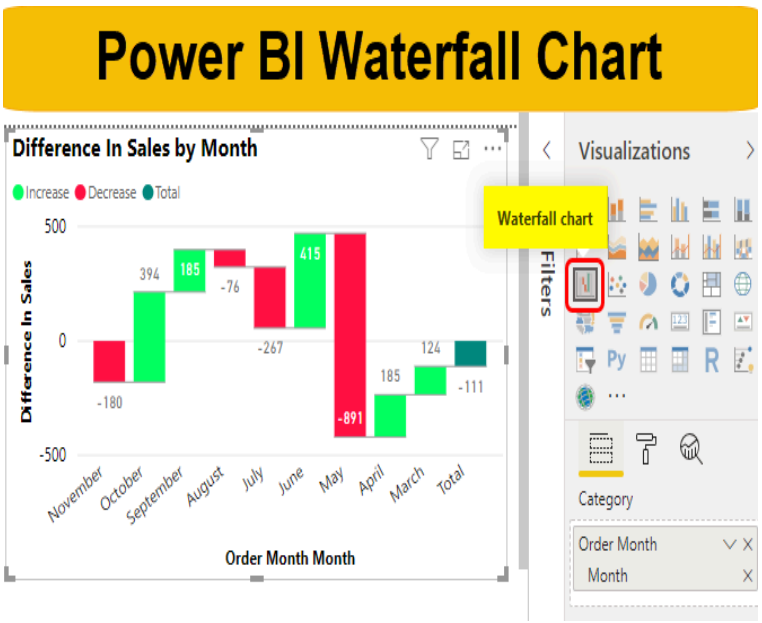
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6.4 Drill Down

You can drill down:

- From Category to Subcategory.
- From Subcategory to Product Level (if defined).

Waterfall charts visually explain the step-by-step construction of totals.

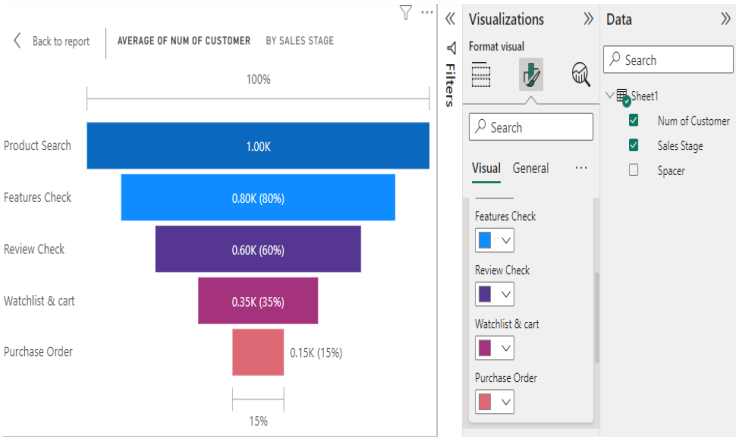


7. FUNNEL CHART

7.1 Purpose

A funnel chart is shaped like a funnel:

- The wider top indicates larger values.
- The narrow bottom indicates smaller values.



7.2 Best for

- Sales pipelines (Leads → Opportunities → Conversions)
- Filtered breakdown of categories
- Sequential processes

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7.3 Creating Funnel Chart

- Use the same fields from the waterfall chart.
- The funnel shape will automatically portray descending magnitude.

7.4 Formatting

- Colors, effects, and labels can be customized.
- Drill-up and drill-down capabilities supported.

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8. SCATTER CHART (SCATTER PLOT)

8.1 Purpose

A scatter chart represents:

- Two continuous variables (X and Y axis)
- Individual data points plotted to identify patterns or correlations.

8.2 Requirements

Unlike bar or column charts:

- Scatter plots require **two numeric variables**.
  Examples:
- X-axis → Sales
- Y-axis → Profit
  Optional:
- Size → Quantity
- Category → Customer/Region

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8.3 Steps to Create

1. Add a scatter chart visual.
2. Drag numeric variables to X and Y.
3. Add a numeric field to "Values" (not used for plotting, but to connect row-level data).
4. Customize markers, labels, gridlines, trendlines.

8.4 Trendline Analysis

- Adding a trendline helps identify the direction of relationship:
    - Upward slope → Positive correlation
    - Downward slope → Negative correlation
    - Flat → No strong relationship

8.5 Interpretation Example

If profit does not strongly increase as sales increase:

- The relationship is weak.
- Indicates variability in cost structures or pricing strategies.

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9. COMMON FORMATTING OPTIONS IN POWER BI VISUALS

Across all visuals:

- **Titles** can be customized.
- **Gridlines** can be enabled/disabled.
- **Data labels** can be adjusted in size, font, and style.
- **Colors** can be modified for clarity.
- **Legends** can be repositioned or replaced with series labels.
- **Zoom sliders** help in focusing on smaller periods.
- **Plot area backgrounds** can be enriched with images or colors.

These formatting controls greatly improve clarity, readability, and professional appearance.

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10. SUMMARY OF ANALYTICAL VALUE

Visualization	Best Used For	Key Insights
Area Chart	Trend comparison of multiple continuous variables	Sales vs Profit movement
Stacked Area	Total + component contribution	Category contribution to overall trend
100% Stacked Area	Share percentages	Proportion dominance
Line + Column Charts	Two different metric types	Volume vs ratio comparisons
Ribbon Chart	Ranking and movement	How ranks change over time
Waterfall Chart	Building totals step-by-step	Contribution analysis
Funnel Chart	Process-based decline	Drop-offs in pipeline
Scatter Plot	Correlation between two variables	Strength & direction of relationship

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11.Conclusion

The collection of visuals covered—area charts, stacked visuals, combo charts, ribbon charts, waterfall charts, funnel charts, and scatter plots—serve a wide spectrum of analytical needs in Power BI. These charts help users interpret trends, compare measures, understand category contributions, observe ranking behavior, navigate through hierarchical data, analyze proportional changes, and evaluate relationships between numerical variables. With the extensive formatting tools available, analysts can customize visuals for clarity, insight generation, and managerial decision-making.